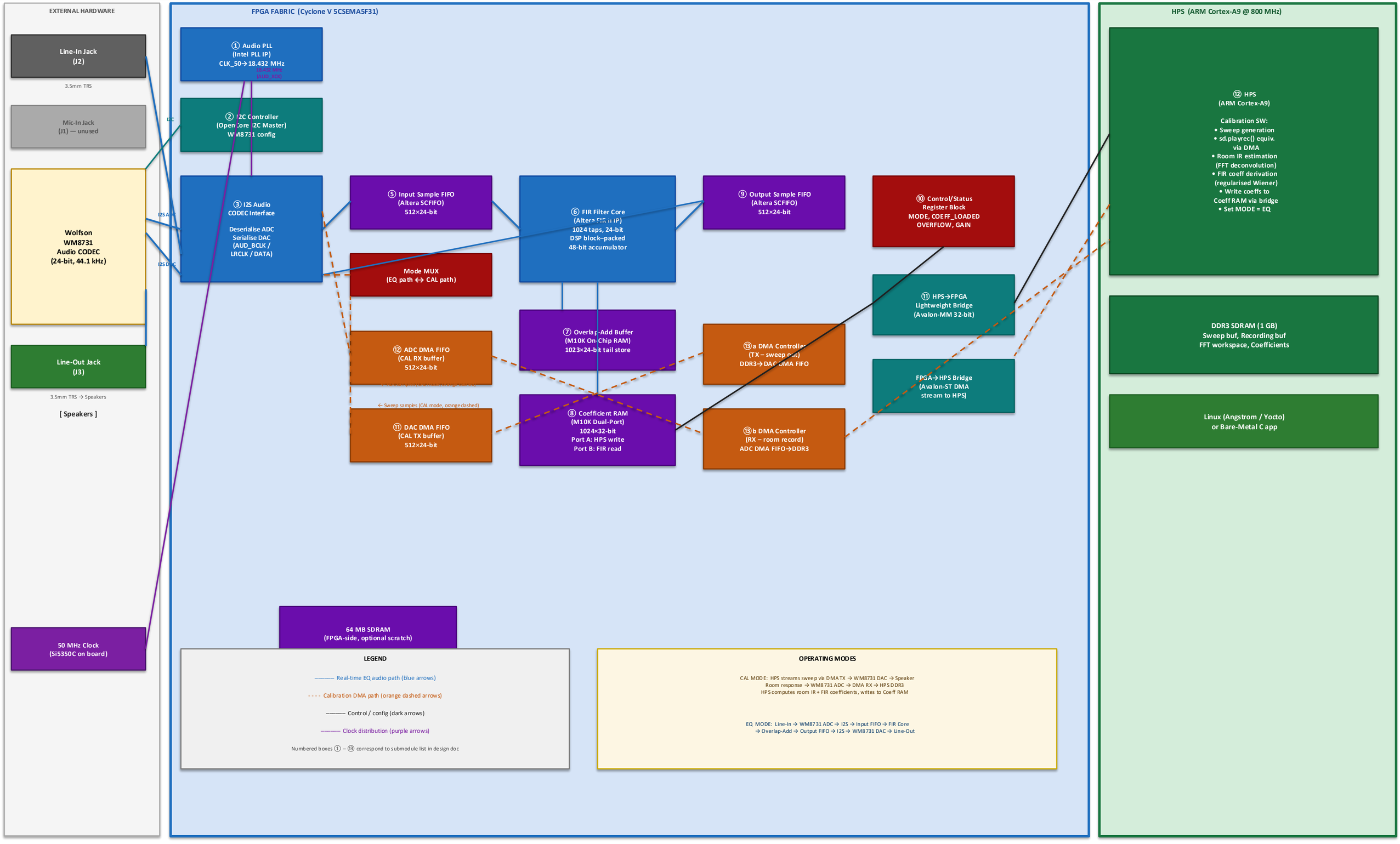DE1-SoC Real-Time Room EQ Correction — System Block Diagram